



Strong culm: a crucial trait for developing next-generation climate-resilient rice lines

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Abstract

Lodging, a phenomenon characterized by the bending or breaking of rice plants, poses substantial constraints on productivity, particularly during the harvesting phase in regions susceptible to strong winds. The rice strong culm trait is influenced by the intricate interplay of genetic, physiological, epigenetic, and environmental factors. Stem architecture, encompassing morphological and anatomical attributes, alongside the composition of both structural and non-structural carbohydrates, emerges as a critical determinant of lodging resistance. The adaptive response of the rice culm to various biotic and abiotic environmental factors further modulates the propensity for lodging. Advancements in next-generation sequencing technologies have expedited the genetic dissection of lodging resistance, enabling the identification of pertinent genes, quantitative trait loci, and novel alleles. Concurrently, contemporary breeding strategies, ranging from biparental approaches to more sophisticated methods such as multi-parent-based breeding, gene pyramiding, genomic selection, genome-wide association studies, and haplotype-based breeding, offer perspectives on the genetic underpinnings of culm strength. This review comprehensively delves into physiological attributes, culm histology, epigenetic determinants, and gene expression profiles associated with lodging resistance, with a specialized focus on leveraging next-generation sequencing for candidate gene discovery.

Keywords Lodging resistance · Climate resilience · Physiological and molecular mechanisms · Breeding efforts · Genes identified

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Abbreviations

SD-1	Semi-Dwarf 1
COVID-19	Corona Virus Disease of 2019
UTM	Universal Testing Machine
FR	Flexural rigidity
YM	Young's modulus
SMI	Secondary moment of inertia
XTR	Xyloglucan endotransglucosylase
SSC	Short and solid culm
<i>APO</i>	Aberrant Panicle Organization
<i>IIP4</i>	ILA1-interacting protein 4
SCM2	Strong Culm 2
<i>Ces</i>	Cellulose synthase
<i>CSL</i>	Cellulose synthase-like
<i>4CL</i>	4-Coumarate ligase
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
<i>CAD</i>	Cinnamyl alcohol dehydrogenase, involved in de novo biosynthesis of pyrimidine nucleotides: glutamine-dependent Carbamoyl phosphate synthetase, Aspartate transcarbamoylase and Dihydroorotase
<i>CalOMT</i>	5-Hydroxyconiferaldehyde O-methyl transferase
<i>bHLH</i>	Helix-loop-helix
<i>GA20ox-2</i>	F GA20 oxidase
<i>COMT</i>	Catechol-O-Methyl Transferase
<i>TUT1</i>	Terminal Uridyl Transferase 1
<i>OsCCC1</i>	Cation chloride cotransporter
SCM3	Strong Culm 3
M	Bending moment at breaking
SM	Section modulus
SMI	Section modulus index

Introduction

The discernible rise in temperature, erratic precipitation patterns, and an alarming surge in the intensity of storms and cyclones pose imminent threats to rice cultivation, disrupting traditional farming practices. The cyclonic turbulence has been intensifying day by day and disrupts the onset of monsoons, a pivotal determinant for agricultural success. Hence, it is becoming increasingly important to develop rice varieties having strong culm and lodging resistance along with high yield for sustainable rice production. In the present review we have systematically surveyed the available literature in the context of strong culm including the physiological, biochemical, breeding, biotechnological aspects. The significance of strong culm in the current scenario of climate change for making rice production resilient especially for the coastal regions has been emphasized. The development, physiology and biochemical pathways involved in

the development of strong culm are highlighted. The phenotyping methods deployed for measuring and assessing culm strength have been described. Breeding improvement strategies inclusive of molecular and conventional breeding have been put forward. We have mentioned the gaps in research that need to be addressed using recent approaches and strategies. We have also highlighted the new breeding strategies, modern artificial intelligence-based techniques for accurate phenotyping and the need for integrating such strategies for developing rice lines having strong culm. The present review comprehensively provides a holistic view of the challenges posed by climate change to rice production and offers insights into potential solutions through breeding and biotechnological advancements. It also underscores the importance of ongoing research and the integration of cutting-edge technologies for sustainable rice cultivation.

Importance of strong culm

Strong culm is a characteristic that aids the plant to be lodging resistant and produces a high grain yield under adverse climatic conditions like strong winds and storms. Strong culm rice varieties with well-developed root systems can fill up the grains with good nutrition (Shrestha et al. 2020). Since 2010, there has been a surge in natural disasters and around 63% of worldwide agriculture has been majorly affected in recent decades (Sarkar et al. 2019). The post-COVID times have indeed pushed the need for cyclone-resistant, high-yielding, and biofortified rice grains (Koc and Karayigit 2022). The strong culm trait is also immensely crucial for dry direct-seeded rice cultivation, an alternative economic water-saving method where rice seeds are sown directly into the dry soil with need-based irrigation (Padmashree et al. 2023).

The historical developments that have been achieved in strong culm research in various studies and the phenotyping models that can be implemented in strong culm measurements have been presented in Fig. 1. This review focuses on the techniques that have been used to analyze the strength of the culm, biotic-abiotic environmental factors influencing culm and yield perspectives, and the molecular mechanism responsible for the strong culm. The conventional breeding approaches and marker-assisted selection approaches have also played key roles in determining the lodging resistance traits in rice and have been discussed.

Sd-1 gene relevance to strong culm

The semi-dwarf gene, *sd1* involved in 'Green Revolution' in rice is deficient in the production of Gibberellin (GA), a plant growth hormone. The mutant phenotype of *sd-1* results from a deficiency in active GAs located in the elongating stem that emerges from a loss of function in the *GA20*

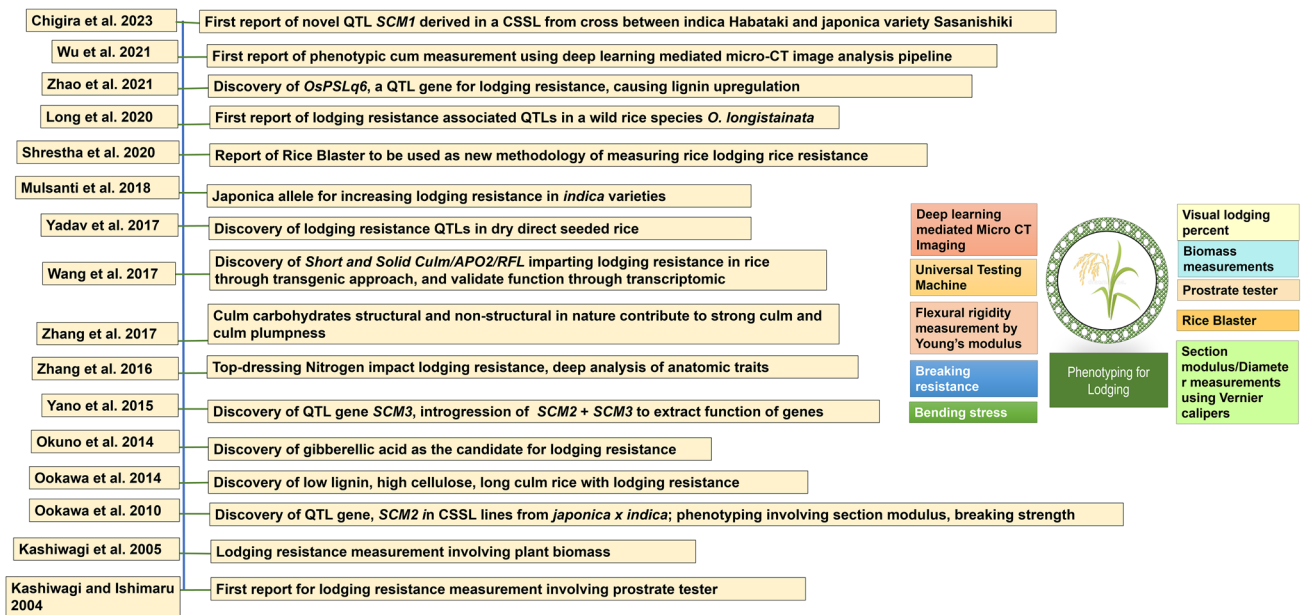


Fig. 1 Historical events in the strong culm research and methods of phenotyping for lodging

oxidase gene (*GA20ox-2*) located in the semi-dwarf-1; *SD-1* locus and is responsible for GA synthesis. The *sd1* alleles were responsible for decreased biomass in lower sections of the culm and hence were unable to support greater growth and yield on the excess application of urea. Recent reports exist that prove the dwarfism that is induced by *sd-1* does not facilitate light penetration in the canopy and leads to reduced biomass production and a lower harvest index (San et al. 2020).

Plant height correlation with yield significance

Employing the long culm cultivars that support high grain yield can increase the productivity of land utilized as explained in the case of ‘Monster Rice’ in China which has been developed with a height of 130 cm, with enhanced yield characters and strong culm (Nomura et al. 2019). The main purpose of the “Monster rice” was that despite height above 130 cm, it had high cellulose content in culms giving increased culm strength. Thus, an increase in the height of the plant with no focus on culm strength or lignin and cellulose partitioning may be detrimental to the grain yield owing to the lodging-related consequences.

An increase in height of the plant leads to greater spacing between the internodes hence greater spacing between leaves that may lead to an efficient distribution of CO₂ and light conditions to facilitate photosynthesis. This facilitates greater assimilation of photosynthates compounded with efficient transport may lead to greater assimilation of starch in the sink tissues and subsequently contribute to good yields (Li et al. 2017). Thick cortical fiber tissue and

well-developed secondary walls mediated by cellulose have been demonstrated to provide great strength to the culm (Ookawa et al. 2010, 2014). Likewise, the use of tall cultivars with good culm wall reinforcement of lignin and cellulose would serve to support increased yields and reduced lodging.

Biochemical components of culm

Lignin

Lignin is a natural phytochemical and an important secondary metabolite that contributes not only to the mechanical stem strength of the rice plant but also aids in the transport of water by reducing the elasticity of xylem vessels and micronutrients and sugar components because of its impermeable nature (Barros et al. 2019). In a study performed on 56 rice varieties in Japan, it was found that lodging resistance is imparted to rice as lignin deposition increases simultaneously (Liu et al. 2018b). Hence lignin production and accumulation in the rice culm cell wall is pivotal for the rice strong culm genotype. Concerning the discoveries made about lignin biosynthesis in rice, common monolignols include coniferyl alcohol-G unit, sinapyl alcohol-S unit, and p-coumarylalcohol-H unit. The G lignin is the most abundant out of the G, S, and H lignin in the rice culms and is responsible for providing mechanical support to the culm in a live plant (Rosado et al. 2021).

In the strong culm rice genotype, 4 Coumarate: CoA ligase suppression leads to lower accumulation of lignin

onto cell walls suggesting its positive role in the phenotype of the plant. Proteomic study reveals 5-Hydroxyconiferaldehyde O-methyl transferase (*OsCaldOMT1*), as a catalyzing enzyme for the reduction of 5-hydroxyconiferaldehyde to sinapylaldehyde that subsequently leads to significantly higher production of S-lignin and converts selgin-lignin to tricin-lignin (Lam et al. 2019). The *gh2* loci in rice variety code for a mutated CAD enzyme, that catalyzes the final step in lignin biosynthesis leading to a lower accumulation of lignin. However, the thick cortical cells have increased cellulose and hemicellulose contents that make the low *gh2* rice strong enough to resist lodging. This suggests a mechanism of suppression of lignin formation to produce cellulose for strong culm rice (Ookawa et al. 2014). Hence in strong culm rice CAD expression is supposed to increase. Moderate nitrogen fertilizer application leads to lignin deposition to the secondary cell walls in rice and helps generate strong culms in rice (Zhang et al. 2017). Application of excess nitrogen was also found to inhibit the Laccase activity that catalyzes the polymerization of monolignol subunits and hence low lignin concentration in rice culms (Schneider et al. 2018).

In addition, there are cellulose and hemicellulose components that also provide strength to the culm at the fruiting stages. Rice Wall Associated Regulators (e.g., *WAP1*, *WACH1*, *WAHL1*, *WADH1*, *OsMYB13a*, and *OsMYB13b*) monitor cell wall-associated lignocellulose content in rice (Zhao et al. 2019). Reduction in cellulose is often countered by increased production of lignin-strong culm rice genotypes (Zhang et al. 2020). Hence adequate light conditions are necessary for the expression of cellulose synthase genes (*OsCesA1*, *OsCesA3*, and *OsCesA8*) and genes related to their deposition onto the primary cell wall (Wu et al. 2017).

Transcription factors regulating cellulose and lignocellulose accumulation

The MYBs (v-myb avian myeloblastosis viral oncogene homolog) belong to a large superfamily of transcription factors, involved in the regulation of secondary cell wall formation. Among the MYBs, *MYB46* and *MYB83*, comprise the second layer of the main switch for secondary wall biosynthesis, and coordination of upstream and downstream transcription factors. It has been reported that *MYB43/52/54* mediates the formation of xylan, *MYB42*, *MYB85*, *MYB58*, and *MYB63* promote lignin formation. *MYB103* promotes cellulose formation, *MYB4* represses lignin formation, and *MYB75* represses cellulose formation. *OsMYB103L* is involved in increased cellulose expression by inducing the CESA genes involved in cellulose synthesis. RNA interference demonstrated that repression of *OsMYB103L* leads to repression of cellulose expression (Xiao et al. 2021; Yang et al. 2014). The *OsNAC* domain-containing transcription

factor, *OsSND2* has been demonstrated to have increased expression of cellulose in culm cells. Knockout experiments involving CRISPR lead to lower cellulose accumulation (Ye et al. 2018). *OsMYB305* has been reported to increase nitrogen uptake and negatively regulate lignocellulose biosynthesis, deposition to rice culm, and enhancement of the concentration of free carbohydrates in culm (Wang et al. 2020). Table 1 provides a list of genes that are involved in the culm growth, development, and other physiological processes.

Hemicellulose

It has been observed that genes controlling hemicellulose biosynthesis were up-regulated (*CSLB01*, *CSLB02*, *CSLB03*, *CSLB04*, *CSLC12*, and *CSLG3*) along with down-regulation of several lignin biosynthetic genes viz. *PAL2*, *PAL3*, and *4CL3* in a coordinated manner at the various growth stages in the plant. Sugar moieties of GDP-Fructose, GDP-Mannose, UDP-Galactose, UDP-Xylose, and UDP-Rhamnose are sugar units essential for hemicellulose synthesis. Cellulose synthase-like genes also form a family of glycosyltransferases and are present in rice namely *CSLA*, *CSLC*, *CSLD*, and *CSLE*. A xylan xylosyltransferase (*XYXT1*) is an essential enzyme recently discovered that specifically adds xylosyl residues to xylan in hemicelluloses thereby strengthening the hemicellulose integrity (Zhong et al. 2018a; b). Overexpression of UDP-glucose galactose epimerase helps in a robust cell wall that subsequently adds to the mechanical strength of the culm as it leads to the ordered arrangement of microfibrils in the cell wall (Liu et al. 2020). The *XYLP* gene code for arabinogalactan protein that is expressed in the culm and is also partly responsible for saccharification of the culm. The *GT61* family glycosyltransferase named *XYXT1* encodes β -1,2 xylosyltransferase protein that is targeted to the Golgi body for xylan biosynthesis and mediates the addition of xylosyl side chains onto xylan and increases the saccharification of the culm (Zhong et al. 2018a).

Silicon, potassium in rice culms

Silicon and potassium play fundamental roles in imparting lodging resistance as it is bound to cellulose, hemicellulose, and lignin in the cell walls. Strong culm genotypes have a higher accumulation of silicon and potassium onto their cell walls forming silicified tissues. An increase in the thickness of lignified dermal sclerenchyma in addition to increments in cross-sectional area and the number of vascular bundles in the lower internode has been documented (Rao and Susmitha 2017; He et al. 2015). It has been found in a study on Japonica rice cultivars that Si and K contents were higher in the heading stage and gradually decreased as it progressed from the heading stage to the dough stage. This in turn leads to decrements in lodging resistance capability

Table 1 Genes that provide resistance to the culm against biotic, abiotic stresses and other physiological processes in the culm and their cross-talk with hormonal pathways

Gene name	Physiological function	Physiological factors	Crosstalk with other factors	Observed effect	References
<i>OsIRT1</i> , <i>OsIRT2</i> , <i>FRO2</i> (<i>Ferric reduction oxidase 2</i>)	Relieves plants of iron toxicity	Plasma membrane proteins regulating iron capture and uptake	Iron plaque formation interferes with iron uptake by channels	Exogenous GA ₃ application relieves stress caused by iron plaque	Mahender et al. 2019
<i>OsPSTOL1</i> (<i>Phosphorus Starvation Tolerance</i>)	Altered root architecture and root surface area to overcome phosphorus starvation tolerance at various stages of growth of rice plant and enhances grain yield under low phosphorus conditions	Serine Threonine kinase activity containing protein, involved in phosphorus transport	Activity of <i>OsPSTOL1</i> may be controlled by <i>OsARF12</i>	<i>ARF12</i> functions in phosphate homeostasis. Mutation in <i>ARF12</i> gene leads to phosphorus toxicity	Chithrameenal et al. 2018
<i>OsNAR2.1</i> (<i>Nitrate assimilation related</i>)	Enhanced nitrogen use efficiency in dry direct seeded conditions and in transgenic experiment	Nitrate uptake transporter, yield improvement	Interacts with <i>OsNIT1</i> and <i>OsNIT2</i> in a synergistic manner	Mediates the IAA ratio of free to conjugate forms and gets transported phloem tissues	Chen et al. 2020
<i>OsSMOS-1</i>	Transcription factor that controls culm growth and cross-sectional diameter	<i>RLA/SMOS-1</i> interacts with <i>OsBZR1</i> to enhance the transcriptional activity and regulates Brassinosteroid signaling and plant architecture	Phosphorylation by <i>GSK2</i> leads to reduced activity of <i>RLA/SMOS-1</i>	Phosphorylation by <i>GSK2</i> leads to negative regulation of Brassinosteroid signaling	Qiao et al. 2017
<i>Pi54</i> , <i>Pi56</i> , <i>Pizt</i> , <i>Pi9</i> , <i>Pi2</i> , <i>Pigm</i>	Zinc finger, LRR domain-containing protein that activates provides resistance against geographically divergent strains of Magnaporthe oryzae	Triggers defense response against rice blast disease resistance and also provides broad spectrum resistance Transgenic rice lines of <i>Pi54</i> demonstrate increased expression of laccase, cellulase, PAL, peroxidases, NAC domain TFs, MADs box, Zinc finger, bZIP and WRKY genes	<i>Xoo1488</i> , <i>Xoo2875</i> effectors inhibits <i>Pi54</i>	<i>Xoo1488</i> , <i>Xoo2875</i> targets <i>OsSERK1</i> activity of trans-auto phosphorylation of <i>OsBRI1</i> thereby inhibits Brassinosteroid mediated immunity and rice resistance to <i>X. oryzae</i> pv. <i>Oryzae</i>	Singh et al. 2020
<i>Xa21</i>	Pattern recognition receptors mediating rice plant immunity	<i>Xa21</i> -recognizes protein called <i>RaxX-sY</i> secreted by <i>Xoo</i> and leads to triggered immune responses	<i>Xb24</i> inactivates <i>Xa21</i> mediated immunity	<i>Xb24</i> when overexpressed physically associates with <i>Xa21</i> leads to decreased <i>Xa-21</i> mediated immunity. Immunity requires dissociation of <i>Xb24</i> from <i>Xa21</i>	Chen et al. 2021

in the rice culm from the heading stage to the dough stage. The optimum application of K as fertilizer positively correlates with enhanced lignin deposition to vascular bundles and sclerenchyma cells leading to increased culm thickening and enhanced lodging resistance in rice (Zaman et al. 2015).

Non-structural carbohydrates

The non-structural carbohydrates are also one of the abundant constituents of the rice culm during the reproductive stage transitions. The bulk of the components are glucose, fructose, sucrose, and starch (Wakabayashi et al. 2022). These non-structural carbohydrates also impart biomass to the rice culms that subsequently impart rigidity to stem cells in rice. Sucrose, starch formed from glucose and fructose, was found in significant amounts in rice culm. Shading influences the accumulation of non-structural carbohydrates negatively and thereby influences culm characters that further reduce grain yield (Wu et al. 2017). Starch accumulates in the rice culms at the maturity stage in various rice cultivars increasing culm plumpness. The amounts of glucose and fructose, also known as soft carbohydrates, obtained at maturity in the culms of cultivar Habataki, reported for carrying *SCM2* is 43.5% higher in Habataki and 34.1% higher in cultivar Leaf Star, compared to Koshihikari. Leaf Star variety has also been reported as carrying low lignin and high cellulose-containing lodging tolerant rice and is a notable mention towards the production of semi-tall lodging tolerant rice germplasm in Japan (Minghua et al. 2021; Zhang et al. 2017).

Physiology

Role of hormones in lodging resistance

Plant hormones play important roles in the modulation of several epigenetic processes and changes in gene expression that help the rice to adapt to changing environmental conditions. Gibberellin (GA) has been a key hormone that controls the breaking type lodging resistance in rice. This is due to the increased accumulation of lignin in their culms and the increased diameter of the culms. In addition to increasing the width of culms, GA also affected an increase in plant height by controlling helix-loop-helix transcription factors up to 120 cm in a variety of T-65 (Nomura et al. 2019). From plant senescence to embryogenesis, auxin signaling is essential for several developmental processes. Auxins regulate apical dominance and are involved in the accumulation of other hormones. During tiller growth, it induces the accumulation of auxin and cytokinin and inhibits the tiller bud outgrowth. In addition, auxins regulate cell elongation and divisions in the apical region of the stem and influence the

grain size. It has been found that *OsIAA10* phosphorylation helps in potentiating *OsARF4-mediated* auxin signaling that helps regulate the grain size and shape. A transcription factor *TGW3* inhibits grain size, and the *OsIAA10* phosphorylation of *TGW3* leads to positive regulation of grain size. Auxin further inhibits tiller bud outgrowth by downregulation of cytokinin biosynthesis in the roots (Ma et al. 2023). Cytokinin regulates the number of branches in a spikelet and influences the thickening of culm diameter and cortical fiber tissue thickening by the mediation of *SCM1* (Strong culm 1). Overall cytokinin works alongside auxins and regulates culm thickening so that it might support additional grain yield without culm breaking (Ookawa et al. 2022).

Gibberellin in conjugation with brassinosteroid causes upregulation of *OsXTR1*, *OsXTR2*, *OsXTR3*, and *OsXTR4*, xyloglucan transferase enzymes responsible for loosening cell walls causing cell elongation in the elongating zone of the internodes and nodes. Genetic mutants carrying abnormally tall phenotypes show a positive correlation with the expression of *OsXTR1* and *OsXTR3*. The QTL *SCM3* contains the TEOSINTE BRANCHED1 (*OsTB1*)/FINE CULM1 gene. The *OsTB1* codes for a *TCP* family transcription factor that acts downstream of strigolactones and works to negatively regulate tiller branching and controls the plant architecture. Experiments have demonstrated that the strigolactone-driven *SCM3* gene actively increases the cross-sectional diameter of the stem in rice thereby increasing the culm strength and simultaneously positively regulating panicle branching and tillering ability (Yano et al. 2014). Measurements of lodging in rice have been demonstrated in Fig. 2.

Brassinosteroids are known to control plant architecture and grain filling. In addition, it has been found that strigolactones and brassinosteroids antagonistically regulate the stability of *D53* and/or *OsBZR1-RLA1-DLT* module in the brassinosteroid signaling pathway to determine rice tillering via *FC1* expression (Fang et al. 2020). The *OsGRAS19* has been reported to positively regulate the brassinosteroid signaling. A recent report depicts *OsGRAS19* knockdown mutant at the *d26* locus mediated by CRISPR/Cas9 positively regulates grain shape but plays negatively in maintaining the height of the rice plant with greater culm cross-sections at the basal internodes (Lin et al. 2019). An experiment on the overexpression of cytokinin response regulator (*OsORR2*) was found to influence rice height at the reproductive stage. Hence, the relationship between the culm strength with that of cytokinin may also be tightly regulated (Shi et al. 2020).

The development of the rice culm and the extent of lodging resistance developed is a complex affair mediated by the regulation of a wide array of pathways related to the hormones are presented in Table 2. The genes associated with the strong culm phenotype are represented in the form of Fig. 3.

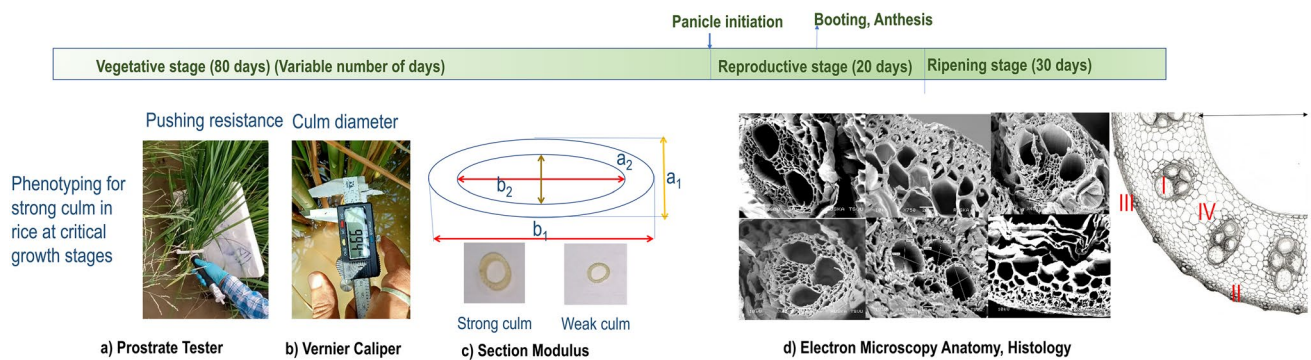


Fig. 2 Widely deployed phenotyping methods for measuring lodging in rice. **a** Measurement using Prostrate tester for measuring culm bending, **b** measurement of culm diameter using Vernier caliper of the main tiller, **c** section modulus of the strong and weak culm **d** electron microscopy of the culm cross-section Section modulus (SM) is calculated as $\pi/32 * (a_1^3 b_1 - a_2^3 b_2)/a_1$, where a_1 is the outer diameter of the minor axis in an oval cross-section, b_1 is the outer diameter

of the major axis in an oval cross-section, a_2 is the inner diameter of the minor axis in an oval cross-section and b_2 is the inner diameter of the major axis in an oval cross-section, I. Large vascular bundles surrounded by lignin-containing bundle sheath cells, II. Small vascular bundles at the outer sclerenchyma layer, III. Lignified sclerenchyma, IV. Parenchyma cell walls with higher lignocellulose reinforcement and higher concentration of structural carbohydrates lignin-containing

Physical parameters to measure culm strength

The bending moment at breaking (M) for the culm is directly proportional to compression/tension at the flexural point/region that leads to breaking/buckling of the point/region in the culm. Section modulus (SM) of a circular section is the ratio of the moment of inertia of the culm cross-section about its neutral axis to the distance of the extreme fiber point from the neutral perpendicular axis. Section modulus (SM) for a culm cross-section is given by: $SM = (\pi/32) \times (a_1^3 b_1 - a_2^3 b_2)/a_1$, wherein a_1 is defined by the outer diameter of the minor axis in the oval cross-section, b_1 is defined by the outer diameter of the major axis in the oval cross-section, a_2 is defined by inner diameter of the minor axis in the oval cross-section, and b_2 is defined by inner diameter of the major axis in the oval cross section.

Bending stress (BS)

Using the Universal Testing Machine (UTM), bending stress (g/mm^2) = bending moment at breaking (gcm)/cross-section modulus (mm^3) while the bending moment at breaking = breaking load $\times$ distance between fulcras (of UTM) $\times$ 1/4. Increment in thickness of culm leads to reduction of bending stress and subsequent increase in lodging resistance. It is dependent on amount of lignin and cellulose in the culm (Mulsanti et al. 2018).

Flexural rigidity (FR)

A bending-type lodging offered by a tiller while encountering bending. It is dependent on Young's modulus also referred to as elastic modulus. It is a parameter used for the

measurement of the stiffness of the rice culm and the lesser it will deform on exposure to strong winds or high grain weight. The culm is a hollow cylinder with an outer diameter surrounding a lamellar tissue followed by an inner diameter wall. $FR = YM \times SMI$; $SMI = SM \times a_1 \times 1/2$, where a_1 is the outer diameter of the minor axis in an oval cross-section; $YM = \text{Young's modulus}$, $SMI = \text{Section modulus index}$. $YM = E = PL^3/(48 \times \delta \times SMI)$ where P is the load applied in the UTM machine, L is the distance between two metal supports in the UTM, and δ is the deflection at the center of the internode (mm) (Mulsanti et al. 2018).

Using a prostrate tester

The prostrate tester was placed in a perpendicular orientation to the stem of the plant and bent at 45° and the reading was noted. The formula is given as culm strength (gram/stem) = [(test reading/40)/no. of tillers] $\times$ 1000 (Yadav et al. 2017). The method is advantageous as the culm samples may be harvested without any tissue damage for downstream applications like next generation sequencing technologies, whole genome sequencing, transcriptomics.

Measurement using a rice Blaster

Artificial storm-inducing machinery is utilized and crops are made to lodge by blasting wind and lodging % is measured in term of stem-base bending moment at breakage. The rice Blaster uses a wind turbine at the plot field scale and create a 1m wide and 6m deep wind path with wind speed adjustable to 30, 45 and 60 km h^{-1} (Shrestha et al. 2020).

Table 2 Various factors governing specific hormonal interactions, and mechanisms involved in the control of rice culm development, growth, and yield

S. no.	Pathway related Hormone	Gene name	Protein domain	Effect on plant	Mechanism of action	References
1	Gibberellic Acid signal transduction	<i>ACCELERATOR OF INTERNODE ELONGATION 1 (ACE1)</i>	Unknown	Accelerates internode elongation, confers cells of intercalary meristem competence for cell division	The combined action of <i>ACE1</i> and <i>DECI</i> determines the domestication phenotype	Nagai et al. 2020
2	Gibberellic acid Signal transduction	<i>DECELERATOR OF INTERNODE ELONGATION 1 (DECI)</i>	Zinc-finger transcription factor	Suppresses internode elongation		Nagai et al. 2020
3	Gibberellic acid Signal transduction	<i>OsMADS7</i>	Transcription factor	Controls plant height, internode elongation and panicle exertion in rice	Binds CArG-box motifs and controls transcription of <i>OsGA2ox3</i> gene	Chu et al. 2019
4	Partial involvement of Gibberellic acid pathway	<i>OsDRP2B/DBC1</i>	Dynamain	Regulates plant height by regulating cell division	Binds DNA Element ACCAAC TTGAA in the promoter of entkaurene oxidase (KO2)	Sang et al. 2020
5	Brassinosteroid signaling	<i>OsGRAS19</i>	Transcription factor	Regulates the height of rice plant and maintains greater culm cross-sections at the basal internodes	Binds in nuclear receptor	Lin et al. 2019
6	Cytokinin Response Regulator	<i>OsORR2</i>	Transcription factor	Controls height of the plant at reproductive stage, overexpression leads to reduced height	Controls gene expression in nucleus by binding through GARP domain, may influence lodging resistance	Shi et al. 2020
7	Gibberellic acid signaling	<i>OsHHLH073</i>	Helix loop Helix transcription factor	Controls height of the plant at reproductive stage and panicle exertion	Possibly binds promoter of KO2 and KO1	Wei and Chen 2018
8	Suppresses Jasmonic Acid responses and promotes Gibberellic Acid Signals	<i>Jasmonate Zim-Domain Protein 9-Slender Rice 1 interaction</i>	Performs Protein–protein interactions	Controls rice growth and Yield	Regulates both JA and GA responses in rice	Um et al. 2018
9	Negative feedback Regulation of gibberellic acid	<i>OsWOX3A</i>	WUSCHEL-related homeobox nuclear protein	Maintains threshold level of endogenous Gibberellic acid Intermediates throughout development	Interacts with KAO promoter and repress KAO expression	Cho et al. 2016

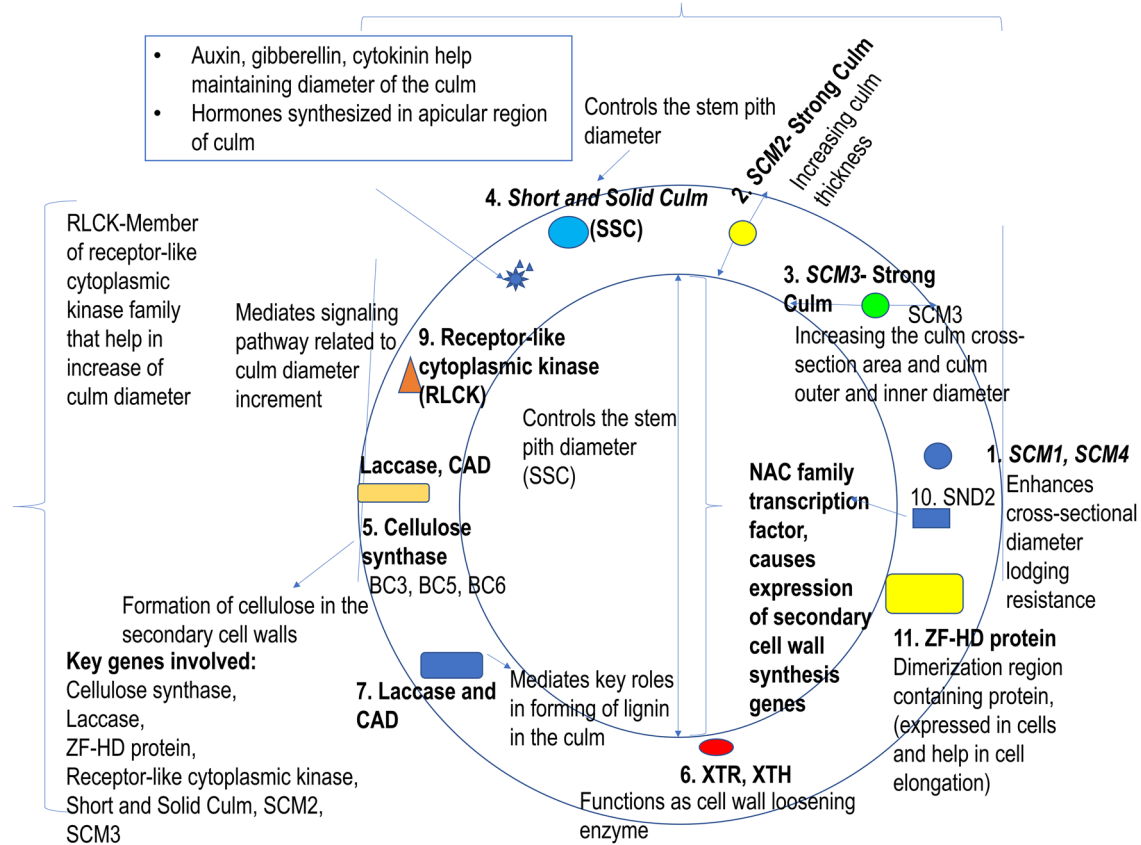


Fig. 3 Genes and pathways associated with strong culm phenotypes. 1. *SCM1*, *SCM4*—Enhances cross-sectional diameter and lodging resistance, 2. *SCM2*—Increase in culm diameter and section modulus and causes increased yield, 3. *SCM3*—Increase in cell size, section modulus, prevents auxiliary tiller formation, 4. *SSC*—Controls culm elongation, pith cavity diameter, mediates GA signaling through KNOX pathway, 5. *BC3*, *BC5*, *BC6*—Cellulose synthesis and deposition, causes cellulose accumulation to cell walls, 6. *XTR*,

XTH—Functions as cell wall loosening enzyme and in cell expansion, 7. *Laccase*, *CAD*—Lignin synthesizing pathway enzymes, 8. *CES*, *CSL*—Cellulose and hemicellulose synthesizing enzymes causing reinforcement of primary and secondary cell walls, 9. *RLCK*—Receptor cytoplasmic kinase, causes effector genes to increase culm strength and lodging resistance, 10. *SND2*—NAC family transcription factor, causes expression of secondary cell wall synthesis genes, 11. *ZF-HD* protein dimerization region containing protein

Table 3 Factors controlling culm strength

S. no.	Factors
1	Bending stress $\propto$ Bending moment at breaking
2	Flexural rigidity $\propto$ Culm strength
3	Cross-section modulus $\propto$ 1/Bending stress
4	Pushing resistance $\propto$ Culm strength

High throughput phenotyping

High throughput micro-CT-RGB imaging system and deep learning (SeqNet) to develop high throughput micro-CT image analysis pipeline for extraction of 24 rice culm morphological traits related to morphology. This technique is accurate and utilizes 4.6 min/plant (Wu et al. 2021). The association of various physical measurements concerning strong culm is presented in Table 3.

Molecular basis of strong culm

The strong culm trait in rice is polygenic and various genes/SNPs are involved in the trait development. Various transcription factors that control the secondary cell wall biosynthesis such as *OsMYB55/61* or *OsBLH6* have been reported. The *IIP4* protein negatively modulates *MYB61*, *CESAs*, and *NAC29/NAC31* function and represses secondary cell wall synthesis. Mutations in the *CCCH*-Tandem Zinc Finger repeat containing *IIP4* protein led to enhanced secondary cell wall synthesis features and enabled stepping up mechanical strength in rice (Zhang et al. 2017). Similarly, *SPL14* is a transcriptional activator that is involved in yield enhancement. *SPL14* gene expression was enhanced under low nitrogen conditions and contributed to the yield in plants (Srikanth et al. 2016). A basic helix loop helix transcription factor protein *OsHHLH073* maintains plant height with changes in gibberellin hormone levels in plant internodes.

Overexpression of *OsHHLH073* causes a reduction in plant height followed by incomplete panicle exertion and lodging-prone cultivars because of less reinforcement of structural carbohydrates onto the culms. Details of studied genes and QTLs that are involved in the formation of the strong culm traits are represented in Table 4.

Next-generation sequencing (NGS) techniques play a crucial role in unraveling the genomic intricacies associated with the strong culm trait in rice. Leveraging NGS allows for a comprehensive exploration of the genetic landscape, identification of key genomic elements, and understanding the molecular basis of robust culm characteristics. The latest developments in next-generation sequencing techniques have paced up the process of investigation into the genome of various tested lodging-resistant rice varieties for many critical mysteries in terms of causal SNP identification (Cheon et al. 2020).

Epigenetics of strong culm

Epigenetics plays an important role in the growth and development of rice. Epigenetic modifications, encompassing heritable alterations in gene function without changes in the DNA sequence, are underscored as pivotal contributors to the regulation of culm endurance in rice. The availability of rice genetic information is key to developing lodging-resistant cultivars. Epigenetic modifications such as methylation occur at CG, CHG or CHH (H = A, C or T) sequences (Chu et al. 2019). In plants, cytosine residues are methylated in CG and non-CG (CHG and CHH), with non-CG methylation being predominant in transcriptional elements and CG methylation occurring in transposable elements and genes. Modulation of gene expression by DNA methylation and repression of transcription by transgene silencing, genome imprinting, and stability of the genome helps regulate various traits in plants (Deng et al. 2017; Wang et al. 2016). In the super rice, Yongyou12 (*YY12*) tandem repeat array upstream of *IPA1* led to alleviation of epigenetic repression of *IPA1/SPL14* encoded by quantitative trait locus WEALTHY FARMER'S PANICLE led to superior yield and culm thickness features (Zhang et al. 2017). Introgression of a functional epigenetic allele of *IPA1/WFP/OsSPL14* improved grain yield traits and improved culm characteristics in elite indica cultivars (Kim et al. 2018).

The miRNAs are epigenetic modulators, affecting protein levels of the target mRNAs without modifying the gene sequences. Rice architecture is controlled by the interaction between *SPL14* (SQUAMOSA PROMOTER BINDING PROTEIN-LIKE 14) and *miR156*. A transcriptional activator, *SPL14* is involved in the yield enhancement of rice (Srikanth et al. 2016). The *SPL14* binds to

the *SBP* box target motif GTAC and controls the expression of the genes in the biosynthesis of lignin and cellulose in rice culm in indica rice as reported by performing transcriptomic approaches. The *SPL14-mediated* reinforcement of culm cell walls leads to lodging resistance accompanied by high yield parameters. Overexpression of the *miR156/SPL* module leads to the production of rice with a significant increase in culm lignin, cellulose, and high-grain quality characters (Lian et al. 2020). Cleavage of *OsSPL14* by *Os-miR156* if impaired by point mutations in the *OsSPL14* sequence produces ideal plant architecture with strong culms and more panicle branching (Deng et al. 2017). The *Osa-miR535* targets the *SPL4* gene and impairs the blast disease resistance mechanism in rice. It is postulated that *OsSPL4* binds to the promoter of *GH3.2*, an indole-3-acetic acid-amido synthetase, thereby promoting its expression (Zhang et al. 2022).

The microRNA, *OsmiR397* disruption plays a significant role in improving the lignin content of rice culms. The overexpression of *miR397* leads to decreased lignification in cell walls thus promoting the de-domestication phenotype as observed in the case of wild rice (Swetha et al. 2018). Another microRNA, *miR166* engages in post-transcriptional gene silencing of rice Class III HD-Zip genes (*LOC_Os03g43930* for *OsHB5*) responsible for auxin signal from regulation of leaf and shoot development and immunity enhancement against pathogens (Salvador-Guirao et al. 2018). The long non-coding RNAs (lncRNAs) form an important domain in rice development response to various stress responses. A summary of lncRNAs along with their interaction patterns is presented in Table 5.

Breeding-related approaches

Rigorous phenotyping for the compact nature of the plant tiller angles in the resultant population would be a favorable trait for agronomic interest for breeders. Crop breeding often converts into a phenotypic expression of traits of agronomical importance as it is in the case of strong culm in rice and would bear fruit for a more climate-resilient genetic base of genotypes. The incidence of cyclones is expected to increase in the coming years owing to global climate shifts (Chen et al. 2021). Hence it is required to change gears towards the production of biotic and abiotic stress-resistant strong culm cultivars with high-yielding characters to improve food security. Various types of breeding approaches are being practiced worldwide to mitigate the problem of lodging resistance viz. i) Conventional breeding ii) Marker-assisted backcross breeding/Marker-assisted selection and iii) Mutation breeding.

Table 4 Genes, QTLs and transcription factors related to culm strength and yield

S. no.	Name of gene	Function in culm	Coding for	References
<i>Functional genes/QTLs</i>				
1	<i>SCM2 (STRONG CULM 2)</i>	The candidate gene for strong culm in rice, causes an increase in the diameter of the culm and yield	F box protein	Ookawa et al. 2010
2	<i>SCM3 (STRONG CULM 3)</i>	QTL gene that causes an increase in cross-sectional inner and outer diameter causes increment in cell size in the parenchyma and results in higher section modulus, prevents tillering in plants by inhibiting the auxiliary buds of plants, negatively regulates lateral branching	TCP domain containing protein, acts in the downstream of Strigolactone signaling	Yano et al. 2014
3	<i>SSC (SHORT AND SOLID CULM)</i>	Controls the culm elongation through GA signaling and medullary cavity formation in the culm Regulates meristem activity of the culm via KNOX pathway	Contains LFY element binding region	Wang et al. 2017
4	<i>SCM1 (STRONG CULM 1)</i>	Improves culm strength and possibly enhances stem cross-sectional area enhances lodging resistance of the culm	Possibly a MYB family transcription factor	Cui et al. 2020; Golfier et al. 2019
5	<i>STRONG CULM 4 (SCM4)</i>	Enhances lodging resistance of the culm	Strong culm QTL gene	Shah et al. 2019
6	<i>Fine culm 1 (FC1)/OsTBI/SCM3</i>	Prevents tillering in plants by inhibiting the auxiliary buds. Negatively regulates lateral branching, gene identical to <i>SCM3</i>	Works downstream of strigolactone signaling, Transcription factor of TCP family	Minakuchi et al. 2010; Ookawa et al. 2010
7	Shortened Basal Internodes	Causes shortened basal internodes codes for Gibberellin 2-Oxidase causes high yield	Encodes for a key gene associated with Gibberellin biosynthetic pathway	Liu et al. 2018a
<i>Enzymes contributing to structural integrity of the culm</i>				
8	<i>OsCES</i> and <i>OsCSL</i> Glycoside hydrolases and Sucrose synthases	<i>CESA</i> -Responsible for cellulose microfibrils synthesis in cell walls <i>CSL</i> gene family synthesizes hemicelluloses imparts for cell wall integrity in rice culm and prevents lodging	<i>CESA</i> codes for cellulose synthases <i>CSL</i> codes for Cellulose synthase-like proteins	Wang et al. 2016
9	<i>OsXTR1, OsXTR3, OsXTR2, OsXTR4; OsXTH</i>	<i>OsXTR1</i> and <i>OsXTR3</i> gets expressed in the elongating cells of internodes <i>OsXTR2</i> and <i>OsXTR4</i> gets expressed in the nodes and in the divisional and elongating zones of internodes <i>OsXTH</i> and <i>OsXTRs</i> function as cell wall loosening enzymes	Codes for Xyloglucan endotrans-glycosylase, Codes for cell wall expansion enzymes	Uozu et al. 2000; Jan et al. 2004
10	Flexible culm I	Responsible for controlling mechanical strength in rice	Encodes for CAD	Li et al. 2009
11	<i>OsPSL5g6/OsCCR28</i>	Involved in lignin biosynthetic pathway	Codes for Cinnamoyl-CoA reductase	Zhao et al. 2021, Zhao et al. 2019

Table 4 (continued)

S. no.	Name of gene	Function in culm	Coding for	References
12	Laccase (<i>OsLAC</i>)/Peroxidase (POD)	Responsible for polymerization of lignin monomers expression strongly correlated with lodging resistance Increase in concentration without subsequent increase in substrates may lead to lignin degradation	Encodes for Laccase enzymes	Shah et al. 2019
13	Cinnamoyl alcohol dehydrogenase (<i>OsCAD</i>), Quercetin 3-O-methyltransferase 1, Flavonol 3-O-methyltransferase 1, caffeic acid 3-O-methyltransferase, Flavonol 3-O-methyltransferase 1, caffeic acid O-methyltransferase 1, 5-H' coniferaldehyde O-methyltransferase 1, 5-5H' coniferaldehydeOMT1, 5-HCAld OMT1, Sinapyl alcohol dehydrogenase, Protein GOLD HULL AND INTERNODE 2	Responsible for synthesis of lignin monomers essential for culm strength in rice	Codes for <i>OsCAD</i> genes, methyl transferases of lignin biosynthesis pathway	Park et al. 2018
14	<i>OsRLCK191</i>	Candidate gene for improving culm strength and lodging resistance in japonica cultivars	Codes for RECEPTOR LIKE CYTOPLASMIC KINASE 191	Chigira et al. 2020
<i>Transcription factors</i>				
15	<i>OsMYB103L</i>	Controls mechanical strength in rice	R2R3-MYB transcription factor	He et al. 2020
16	<i>OsSND2/OsNAC</i>	Responsible for secondary cell wall biosynthesis	NAC family transcription factor, expressed in internodes and panicles	Ye et al. 2018
17	<i>IPA1</i>	Yield and culm character enhancing gene	Squamosa promoter-binding-like transcription activator	Zhang et al. 2017

Table 5 Summary of all the long non-coding RNA interaction with interacting partners to bring about changes in internode length and also regulate internode jointing stage of the plant

Long non-coding RNA	Interacting partner (Target)	Gene function	Effect on plant	References
<i>LTCONS_00063919</i>	<i>D35/OsKO2</i>	Kaurene oxidase	Affects growth and elongation of plant	
<i>LTCONS_00002962</i>	<i>SLR1</i>	GA signal transduction	Overexpression leads to dwarf phenotype in rice, shorter culms	
<i>LTCONS_00033182</i>	<i>GID2</i>	Degrades <i>SLR1</i>	Mutation causes short Internodes phenotype	Li et al. 2021a, b
<i>LTCONS_00025454</i>	<i>OsARF</i>	Reduced response to auxin	Affects plant height	
<i>LTCONS_00034806</i> & <i>LTCONS_00034805</i>	miR172d-5p	May interacts with <i>Phytochrome A</i>	Mediates elongation of stem nodes	
<i>LTCONS_00001063</i> , <i>LTCONS_00034806</i> & <i>LTCONS_00034805</i>	–	–	May regulate, growth process at stem jointing stage	
<i>microRNA528</i>	RED AND FAR-RED INSENSITIVE2 (<i>OsRFL2</i>)	Flowering under long day conditions	Regulates flowering and lodging resistance	Yang et al. 2019

Conventional breeding

The QTL designated as MVQTL4.1 on Chr. 4 for lodging resistance-associated trait was identified using the BC₃F₃ population derived from Moroberekkan and Swarna (Dixit et al. 2015). Analysis of 253 Swarna x Moroberekkan BC₃F₄ generation plants revealed *qCL1.1*, *qCL2.1*, *qCL7.1* for culm length; *qCD1.1*, *qCD2.1*, *qCD7.1* for culm diameter; *qCS1.1*, *qCS2.1*, *qCS2.2*, *qCS2.3*, *qCS6.1* and *qCS6.2* for culm strength. The major candidate genes obtained in the study were Gibberellic acid 3 β hydroxylase and *SCM2* for lodging resistance (Yadav et al. 2017). 128 RILs were developed using Emi-no-kizua with Tomohomani and QTLs were identified on Chr. 4 (*qGA4*) and in Chr. 8 (*qGA8*). The introgression of *qGA* (encoding for gibberellic acid biosynthesis) on the *sd-1* locus conferred lodging resistance and improved grain weight to the superior progeny line (Nagaoka et al. 2019). In a study of 1000 F₂ individuals derived from Zhu1S (female/male) and SV14 (male/female) reciprocal cross for identification of lodging resistance genetic factors. Analysis has been carried out by use of NGS technologies and has been narrowed down to the Shortened Basal Internodes (*SBI*) gene, which encodes for a *Gibberellin 2-oxidase* and is responsible for controlling the elongation of culm basal internodes through deactivation of GA activity. *SBI* has predominantly been expressed in the basal internodes of the culm (Liu et al. 2018a).

Molecular breeding, marker-assisted backcross breeding, marker-assisted selection

Fine mapping of a genomic region in a chromosome segment substitution line (CSSLs) derived from a cross between a

thick culm bearing improved indica variety, Habataki, and recurrent japonica parent Sasanishiki bearing thin culm was executed. Parameters viz. bending moment at breaking at the fourth internode, section modulus (influenced by culm wall thickness and diameter) and bending stress (influenced by structural and nonstructural contents) had been measured. Two genotypes obtained among the CSSLs namely SL403 and SL420 had wider culm diameters at the basal internodes. Mapping revealed SL403 contained genomic segments of Chr. 1 and SL420 contained genomic segments from Chr. 6 (15.8–124.4 cM) of Habataki. Segment present in Chr. 1 involved in increasing culm wall thickness is identified as STRONG CULM 1 (*SCM1*) and Chr. 6 is involved in increasing culm diameter and is identified as STRONG CULM 2 (*SCM2*); its position was found identical to ABERRANT PANICLE ORGANIZATION1 (*APOL1*) gene and had several insertions, deletions, single nucleotide polymorphisms including insertion of a large retrotransposon (9.5 kb approx.) (Ookawa et al. 2014, 2010). Pyramiding multiple strong culm genes in the background of popular mega varieties would enhance the culm strength and thus would be climate resilient.

A study based on CSSLs derived from a cross between Koshihikari (japonica) and Kasalath (indica) revealed QTL for M of upper culm and increased grain filling to be present in Chr. 11 containing the *bsuc11* region. The QTLs obtained were responsible for increasing the physical strength of upper culms and resistance of upper culms to bending-type lodging. A candidate gene, *PRL4* was discovered for the accumulation of non-structural carbohydrates in a set of chromosome segment substitution lines (CSSLs) at the fully ripening stage (Kashiwagi 2022). In another experiment derived from a cross of Takanari x Koshihikari CSSLs

were used for the identification of QTLs related to traits of lodging resistance. QTL for bending stress was identified in Chr. 3, 5, 6, 9, 10, 11, and 12; QTL for Young Modulus was localized in Chr. 2, 10, 11. QTL for cellulose density in stem (hemicellulose, holocellulose, and cellulose are present in Chr. 1, 3, and 5 (Mulsanti et al. 2018).

Hybrid breeding

Recent hybrid breeding techniques involving wide-cross compatible and inter-sub-specific heterosis combined with genome sequencing leads to the discovery of strong culm with added yield and quality traits e.g., *WEALTHY FARMER'S PANICLE* or *IPA1* discovered in japonica rice variety Shaonie Jing (Qian et al. 2016). The partial restorer line of Swarna was crossed with a stable restorer KMR3R possessing *Rf3* and *Rf4* genes and BC₁F₅ and BC₂F₄ populations were developed by marker-assisted back cross-breeding (MABB). The resulting improved lines of Swarna possessing *Rf3* and *Rf4* genes demonstrated higher grain yield in test crosses and complete fertility restoration. The lines were also reported for having strong culm alongside possessing good grain characters (Ponnuswamy et al. 2020).

Inter-species breeding, multi-parental breeding, transgenic breeding

Breeding experiments in wild rice species have been successful in tapping the genetic diversity associated with lodging resistance traits and other important domestication-related traits. A novel interspecies cross between indica reference 93–11 and *Oryza longistaminata* in the BC₂F₂₀ population identified QTLs *qLR1* in Chr. 1 and *qLR8* in Chr. 8. The *qLR1* was responsible for increased stem diameter and stem length. The *qLR8* was responsible for increased M at lower internodes (Long et al. 2020). *Oryza sativa japonica* cultivar 'Hwasengbyeon' was crossed to *O. grandiglumis* as donor parent. The QTLs for internode diameter and plant height were identified in Chr. 1 as the *sd1* mutant 'Reimei' and the region responsible for internode diameter was limited to 1 Mb region in Chr.2 that harbors the *GW2* gene. The introgressed lines demonstrated increased lodging resistance, yield, and disease resistance (Jeon et al. 2018). In another transgenic endeavor, the expression of *OsEXTL* into the genetic background of Zhonghua 11 (*Oryza sativa ssp. japonica*) leads to the formation of lodging-resistant genotypes in the T3 generation (Fan et al. 2017). Transgenic rice plants bearing *IPA1* upstream repeat region of Chinese super rice containing one or three repeats constructs fused to the *CaMV35S* mini promoter-glucuronidase (*GUS*) revealed the repeats do not have enhancer activity but act to epigenetically overcome *IPA1* repression (Zhang et al. 2017).

The strong culm trait is crucial for sustaining high yields and ensuring the stability of rice crops in diverse agroecological environments. The Multi-Parent Advanced Generation Inter-Cross (MAGIC) populations showcase distinctive advantages compared to conventional bi-parental and germplasm populations. The founder parents from diverse clusters can be selected with prior information for deployment in such breeding programs. These advantages include significant levels of genetic recombination, the absence of genetic structure, and heightened genetic and phenotypic diversity. Similarly, implementing genomic selection for the strong culm trait in rice involves leveraging genomic information to predict the breeding value of individuals without the need for direct phenotypic measurements. Genome-wide association mapping (GWAS) is a powerful approach for identifying genetic markers associated with complex traits such as strong culm in rice.

Wild rice species, being the ancestors of domesticated rice, harbor a diverse gene pool that may contain valuable genetic variations contributing to robust culms. Exploring wild rice varieties presents a promising avenue for enhancing the strong culm trait in cultivated rice. By leveraging the novel alleles present in the wild rice pool, researchers can uncover novel genetic elements associated with strong culm traits. Integrating these elements into cultivated rice varieties can contribute to the development of resilient crops that can better withstand the challenges posed by lodging in diverse environmental contexts. The landraces, traditional locally adapted varieties of rice, offer a valuable genetic resource for enhancing the strong culm trait in cultivated rice (Gupte et al. 2023). These varieties have evolved over generations, adapting to specific environmental conditions, and exhibiting traits that may contribute to improved culm strength. Conducting a network-based genomic survey and analysis to uncover genetic variations associated with strong culm traits in wild rice and landraces is essential for preliminary understanding.

Studies on diverse genotypes

Candidate gene analysis in crossing populations

Candidate gene analysis from a handful lodging resistant stabilized rice genotypes viz., ShenNong9903, YanJian218, KongYu131, and ShenNongK33 was carried out. Expression and phenotypic analysis identified *PAL*, *COMT*, *4CL3*, *CAD2*, *CAD7*, and *CCR20* to be candidate genes for lodging resistance (Liu et al. 2018b). All the above genes catalyze key steps in the production of lignin. Studying a rice population of landraces in Japan, identified QTLs associated with strong culm in Chr.2 harboring gene associated with zinc finger homeodomain-containing proteins for regulation of culm diameter. It has been reported that *GRAIN SIZE2* may

also play crucial roles in not only affecting grain size and weight but may also have pleiotropic effects on culm morphogenesis. QTLs related to bending moment were analyzed in the short arm of Chr.3 associated with the RM4108 microsatellite marker. This study involved 168 landraces in 2018 and 316 landraces in 2019 (Nomura et al. 2021). A diverse panel of 550 accessions was evaluated for two consecutive years, and genomes were mapped using high-quality SNPs and QTLs analyzed for strong culm traits. Candidate gene analysis revealed *TUT1*, *OsCCC1*, and seventeen novel candidate genes that are beneficial for leaf type and lodging resistance (Meng et al. 2021).

The molecular mechanisms associated with a strong culm in rice involve a complex interplay of genetic factors that influence the development and structure of the plant's stem. While the specific mechanisms can vary among rice varieties, several key molecular processes contribute to the development of a robust and resilient culm. Wang et al. (2017) demonstrated the role of SHORT and SOLID CULM (SSC) in imparting strength to rice culm. The SSC affected culm elongation by maintaining gibberellin homeostasis, and medullary cavity (MC) formation via the knotted-1-like homeobox (KNOX) pathway. The SSC regulates the levels or activities of gibberellins, contributing to the proper balance between elongation and strength in the culm.

Genetic diversity analysis for strong culm trait in rice

Using recombinant inbred lines of Omachi (a landrace rice variety in Japan) and Koshihikari, QTLs were discovered on Chrs. 3 and 7 viz. *qCD3* and *qCD7-1* for culm diameter at the 5th internode. The F_2 population resulting from a cross between one of the RILs and Koshihikari further confirmed *qCD7-2* QTL. Transcriptome analysis identified lignin metabolizing genes. It was also found that culm diameter pleiotropically increased the number of spikelets per panicle without affecting the days to heading and culm length. This study proved *qCD3*, *qCD7-1*, and *qCD7-2* as candidates for increasing lodging resistance and yield (Chigira et al. 2023). In another study, QTL analysis of the landrace population of 135 temperate japonica cultivars yielded QTLs for culm strength components, QTL for bending stress, and young modulus in Chrs. 5, 6, and 11. The candidate gene for culm strength was found to be *RLCK*, coding for receptor-like cytoplasmic kinase (Chigira et al. 2020).

The QTL, *qSCM4* has been identified recently using an F_2 segregant population, and a recombinant self-incompatible line population from a cross between Shennong265 (SN265) and Lijiangxintuanheigu (LTH) varieties in China. The physical interval was mapped to 58.5 kb inclusive of 11 genes by utilizing the heterozygous derived segregant population, which included 1781 individual plants, and the BC_3F_2 segregant population consisting of 3216 individuals.

The gene *Os04g0615000*, encoding for a functional protein with structural domains of serine and cysteine, was identified as the most reasonable candidate for *qSCM4* by DNA sequencing. A fragment deletion, two base changes in the 3' UTR region, six base changes in the exons, and four base changes in the introns were among the 13 DNA sequence modifications in LTH compared to SN265 in this gene. By increasing stem thickness by 25.3% and stem bending resistance by 20.3%, a near-isogenic line carrying *qSCM4* demonstrated how lodging resistance was increased. Additionally, it was shown that *qSCM4* increased the number of grains per panicle by 14.7%, the primary branch number per panicle by 16.7%, and the secondary branch number per panicle by 9.9%. These observations provide valuable genetic resources for increasing culm strength and climate resilience (Yang et al. 2023).

The yield-related genes viz. *Gn1a*, *SCM2*, and *GW2* were validated in the indica-tropical japonica-derived lines, and lines having all the QTLs were further studied (Mohanty et al. 2016). In another experiment in Japan, the temperate japonica near-isogenic line (NIL) with the introgression of *SCM1* or *SCM2* locus of the indica variety, Habataki, and another NIL with the introgressed *SCM3* or *SCM4* locus of the tropical japonica variety, Chugoku 117, were developed for evaluation of the effects of four strong-culm genes on lodging resistance. Deploying the double, triple, and quadruple combinations obtained from successive crossovers between NIL-SCM1 and NIL-SCM4, pyramiding lines were created. Owing to the pleiotropic effects of these genes, the quadruple pyramiding line (NIL-*SCM1* + *SCM2* + *SCM3* + *SCM4*) had the greatest culm diameter, highest culm strength, and increased spikelet number among the combinations. Due to their additive effect, pyramiding of strong culm genes improved culm thickness, culm strength, and spikelet number significantly. The findings of this study point to the significance of finding the optimal combinations of strong culm gene alleles among natural variations and pyramiding these genes for enhancing high-yielding cultivars with improved culm strength and climate resilience (Ookawa et al. 2022).

Genome editing efforts for strong culm rice

Genome editing techniques, particularly CRISPR/Cas-mediated approaches, offer a precise and targeted means to enhance the strong culm trait in rice. The utilization of these techniques allows for the modification of specific genomic regions associated with culm strength, offering unprecedented control over the genetic makeup of rice plants. CRISPR/Cas9 is revolutionizing the field of functional genomics. Induced mutations in the 5' UTR of *SCM3/OsTB1* in Nipponbare using CRISPR/Cas9 technique employing different sgRNAs integrated into a single plasmid

and transformed into Nipponbare using *Agrobacterium*-mediated transformation. The transformed lines produced different phenotypes in terms of tiller number, culm length, panicle sizes, and yield parameters. It was also reported that deletion in the 5' non-coding region brought about changes in *OsSPL14/IPA1* interaction with the GTAC motif resulting in null phenotypes (Cui et al. 2020). A genome editing experiment with a high throughput CRISPR-genetic screen was also met with success. In this approach, CRISPR-mediated disruption of the high tillering dwarf (*HTD1*) allele in stabilized *O. Sativa* and *O. Glaberrima* emanated improved lodging resistance with reduced plant stature (Lacchini et al. 2020). *GRAIN WIDTH* and *WEIGHT2* (*GW2*) genes encoding a *RING-type E3 ubiquitin ligase*, determine the grain weight in rice. Knock-out of *GW2* mediated by CRISPR/Cas9 genome editing in indica line (var. MTU 1010) displayed a thick aleurone layer with increased protein content and increased accumulation of (Fe, Zn, K, P, Ca) dietary minerals in the rice grain. Moreover, the mutation also improved early growth vigor with improved root, and culm characteristics resulting in improved morphology (Achary and Reddy 2021). Experiments with multiplex CRISPR/Cas9 construct targeting *HTD1*, *GS3*, *GW2*, and *GNIA* lead to significantly improved lodging resistance and seed yield in African *Oryza glaberrima* and *Oryza sativa* landraces (Lacchini et al. 2020). Examples of transgenics and

CRISPR/Cas9 genome editing for improvement of strong culm in rice are presented in Table 6.

A holistic picture of the research developments in strong culm has been presented in Fig. 4.

Advanced technologies for strong culm research in rice

Artificial intelligence

Artificial Intelligence in many ways has facilitated the trickiest road to the demonstration of agriculture (Zhu et al. 2024; Ritharson et al. 2024). The strategies providing productive efficient harvest yield may improve the involvement of Artificial Intelligence-Machine learning algorithms. Agricultural innovation, pest, and disease management in strong culm disease-resistant rice through computer algorithms to further recognize the recognition of infection, engrossing the images of culm, rice plant, or seeds. Even if they provide a greater outlook on seedling health, grain quality, and reproductive stage development is of importance; K-means cluster algorithms, co-occurrence analysis, etc. further improve the overall data points collected. These platforms help identify real-time crop insights, identification of areas that need irrigation and are the future of irrigation (Aggarwal et al. 2022).

Table 6 Transgenic endeavors undertaken for strong culm rice and the diseases that are being afflicted to the culm at various stages of growth

Transgenic/genome editing experiments involving transcription factors	Factor function/nature	Observed effect	References
<i>R2R3</i> MYB expression, <i>OsNAC5</i> , <i>OsTF1L</i> overexpression	Regulates Phenyl ammonia lyase activity (<i>R2R3-MYB</i>); activates <i>CINNAMOYL-CoA REDUCTASE</i> (<i>OsCCR10</i>); HD-Zip transcription factor	Confers brown plant hopper resistance, Improved abiotic stress tolerance due to enhanced lignin synthesis	He et al. 2020
CRISPR/CAS9 repression of <i>OsUXE</i> or <i>OsXAT</i> genes	UDP-xylose epimerase and xylan arabinosyl-transferase	Decreases arabinose in the lignocellulose backbones and improves saccharification of rice straw	Chen et al. 2021
Endo- β -1,4-glucanase expression	Modification of cellulose features	Enhanced biomass saccharification, bioethanol production	Huang et al. 2019
<i>OsPGIP1</i> expression	Polygalacturonase inhibiting protein	Enhanced plant tolerance to biotic stresses and sheath blight	Chen et al. 2021
<i>OsPGIP2</i> expression			
Brittle Culm19 expression	Cellulose Synthase Subunit <i>CESA4</i>	Affects cellulose synthesis, causes dominant Brittle Phenotype	Ma et al. 2021
<i>OsSND2</i> expression	NAC family transcription factor, regulates MYB family members	Master regulator to mediate Secondary Cell Walls biosynthesis	Ye et al. 2018
<i>OsbHLH073</i> and <i>OsSWN2</i> overexpression	Helix-loop-helix transcription factor	Negatively regulates plant height, internode elongation, panicle exertion by regulation of GA biosynthesis involving <i>OsKO1</i> and <i>OsKO2</i> genes	Lee et al. 2020
<i>OsMYB305</i> overexpression	MYB family transcription factor	Nitrate uptake and assimilation and enhances rice growth	Wang et al. 2020
<i>FRAGILE CULM 18</i> expression	UDP-glucuronic acid decarboxylase	Enhanced xylan biosynthesis, cellulose deposition	Ruan et al. 2022

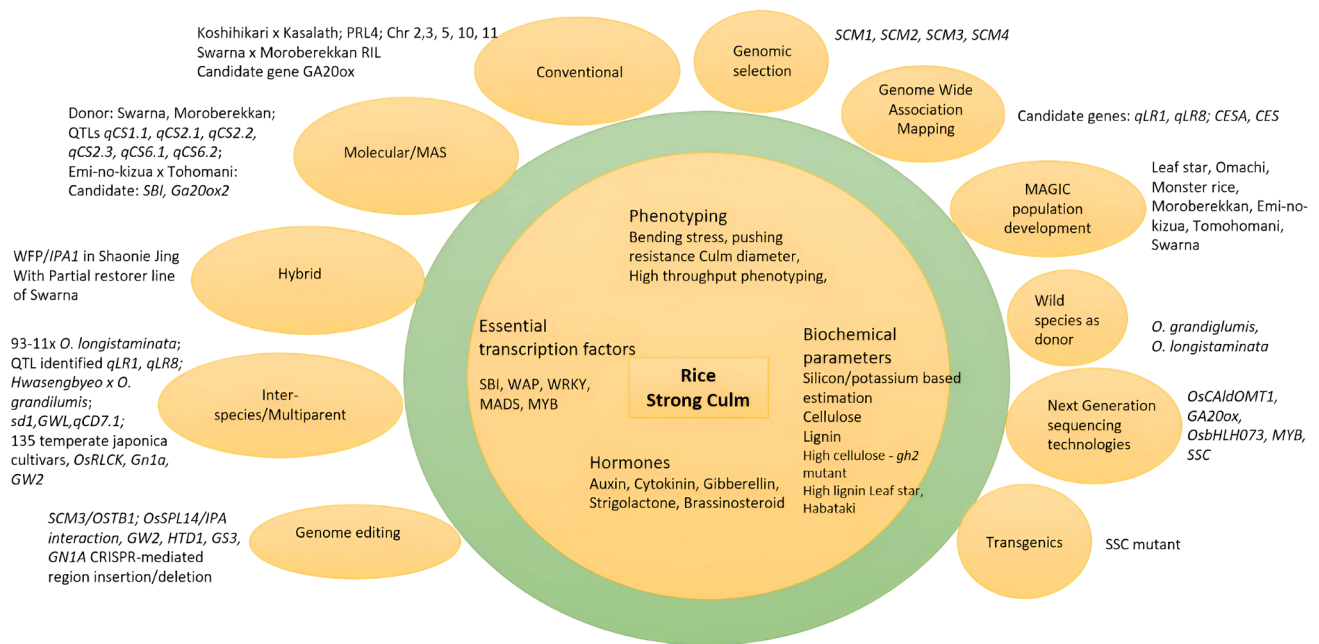


Fig. 4 Overall view of the strong culm research achievements in rice

These technological advances can be very well applied for strong culm research in rice but the factors related to the successful application need to be considered. The AI algorithms heavily rely on large datasets for training and accurate predictions. For specific traits like culm strength, obtaining high-quality, diverse datasets can be challenging. Limited or biased data may lead to suboptimal performance and inaccurate results. Additionally, high initial investment, ethical considerations, cyber security threats, and system malfunctions also need to be considered.

Advanced high throughput phenotyping

High throughput phenotyping platform is a plant-to-sensor type and is designed to identify images from all four sides and top-view of plants (Kim et al. 2020; Li et al. 2021a, b). The high-throughput phenotyping for determining culm strength is required. The HTP platforms are non-destructive and rapid approaches to monitoring and measuring multiple phenotypic traits concerning the growth, yield, and adaptation to biotic or abiotic stress responses. Unmanned aerial vehicles (UAVs) are recently being deployed to predict the stem, leaf weight above ground weight, duration, etc. in rice (Taniguchi et al. 2022). The HTP techniques include spectral, fluorescence, thermal, and three-dimensional imaging for the measurement of the morphological, physiological, and pathological resistance traits in cotton. Various ground-based aerial platforms were also developed to function in the implementation of these HTP systems. Pattern recognition and time series have led to high contributions toward

understanding and monitoring responses of any crop across different environmental conditions or scenarios. The high throughput technique is coupled with advancements in big data accumulation and continuous advances in imaging. Improvements in HTP platforms, for instance, speed breeding in artificially controlled lights in short-day crops, etc. tackle the ability of these systems to capture the variability in agriculture fields will be of great value (Gill et al. 2022; Jahne et al. 2020; Pabuayon et al. 2019).

Speed breeding

Speed breeding involves the rapid generation of plants by reducing their generation time in a controlled environment, unlike the normal glasshouse. The control of the temperature in the full enclosure, controlled-environment growth chambers accelerates the development of plants and is highly deployed for phenotyping of varying plant traits, and forward and reverse genetic approaches (Sharma et al. 2022). Speed breeding has potential in the study of lodging resistance and determination of strong culm rice regarding the development of mapping populations, and molecular breeding. Additionally, the lines can be evaluated and phenotyped for disease tolerance. The generation of the advancement of the work in the shortest possible time will require a price to be paid in terms of infrastructure creation. Improvement and reduction of cost in the aspect of infrastructure development and maintenance is essential for widespread routine adoption of speed breeding (Potts et al. 2023; Xu et al. 2023; Watson et al. 2018).

Recently, a protocol, SpeedFlowr has been established in rice wherein 4 to 5 generations of indica and/or japonica rice can be advanced in a year (Kabade et al. 2023). The speed breeding protocols can expedite the research on strong culm along with high throughput phenotyping techniques for assessing the strength of rice.

Machine learning and deep learning

Digital transformation of agriculture has evolved in various aspects of management into artificial intelligent endeavors for the sake of making value from ever-increasing data originating from various sources. This technology has been deployed pre- and post-harvesting with crop responses to rapid climate changes. Deep learning, being a subset of AI, machine learning in the field of agriculture by a thorough review of the recent scholarly literature-based keywords' combinations of "machine learning" together with "crop management", "water management", "soil management", and "livestock management", and following PRISMA guidelines. The abundance of machine learning algorithms, with those housed by Artificial Neural Networks, are becoming more efficient in excavating deep information regarding crop management. Sensors attached to satellites, unmanned ground, and aerial vehicles detect data and send the information downstream to be analyzed. It is a beneficial technology and many advancements are not yet being made in rice (Attri et al. 2023). Hence, coupling these advanced learning platforms with

high throughput phenotyping has prospective in the near future.

Big data analysis

BIG DATA (BD) and INTERNET OF THINGS (IoT) are expected to have a large impact on Smart Farming that involves the whole supply chain, to rice production. These algorithms have greatly imparted to improve traditional farming in primarily China and followed by the United States, South Korea, Philippines, and other developed countries and nationalities to go from traditional farming to Rice SMART farming. Rice smart farming, data modeling, and deep learning algorithms are to be used for each activity and offer new perspectives towards rice strong culm and climate resilient precision agriculture. The downside of this technology is only cybersecurity breaches and threats and implementation are necessary (Chergui and Khechadi 2022; Alfred et al. 2021). The big data can be deployed in studying various traits including strong culm for climate resilience. An account of these advancements has been presented in Fig. 5.

Conclusion

The robust culm trait in rice stands as a pivotal determinant in fortifying resistance against lodging, particularly under adverse climatic conditions such as storms and cyclones. The expedited pace of genomic exploration, propelled by advancements in next-generation sequencing techniques,

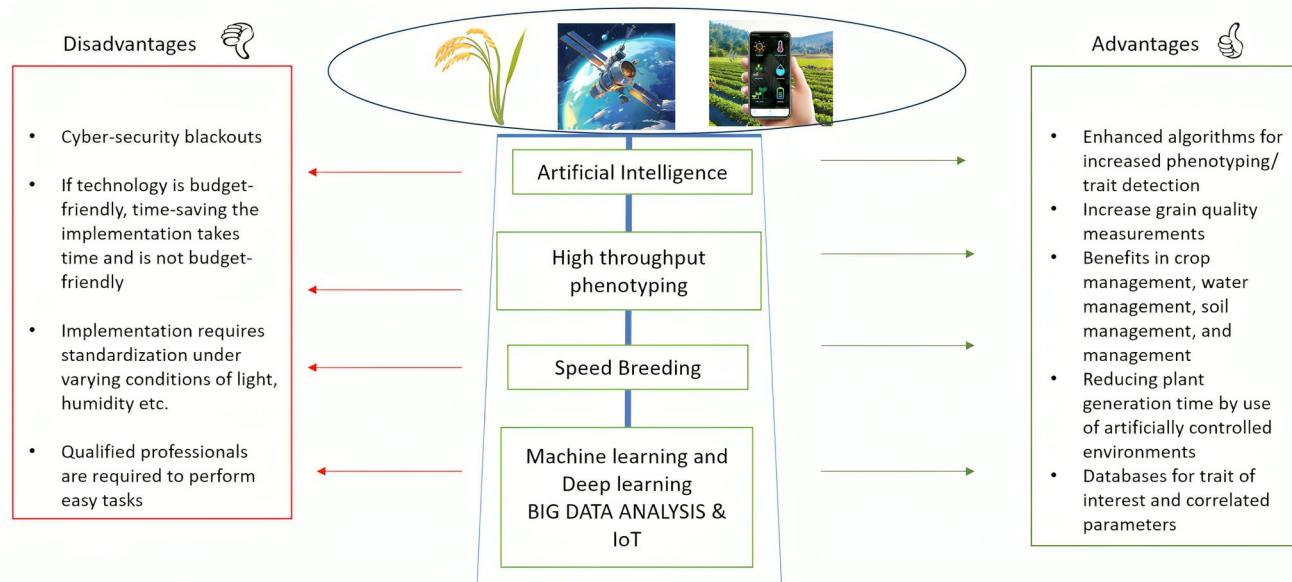


Fig. 5 Cutting-edge next-generation techniques having the potential for assessing the culm strength in rice

significantly augments our understanding of lodging-resistant rice varieties. This progress, particularly in the identification of causal SNPs, not only enriches our knowledge base but also opens new avenues for refining breeding strategies and bolstering rice resilience. Anticipating future advancements, the implementation of CRISPR/Cas-mediated genome editing experiments targeting identified strong culm genomic regions represents an innovative frontier. This precision genetic manipulation approach holds the potential to yield transformative insights, paving the way for the development of rice varieties with finely tuned and optimized strong culm traits. The synergistic integration of traditional, molecular breeding methodologies with state-of-the-art genomic tools holds the potential for the development of rice varieties with finely tuned and optimized strong culm traits. The findings extracted from this review not only summarize the existing knowledge but also outline a definitive path for future research, presenting transformative opportunities in our pursuit of rice varieties resilient enough to withstand the challenges posed by a dynamic climate.

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Declarations

Conflict of interest All the authors mentioned in the manuscript have agreed for authorship, read and approved the manuscript, and given consent for submission. The authors declare that there is no conflict of interest.

Ethical statement The author confirms that no animal or human was harmed.

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